

Valentia Technologies

Corporate website — Technology stack overview

Corporate website for Valentia Technologies (healthcare software, Qatar). A single-page application built with React and TypeScript, featuring smooth scrolling, scroll-driven animation and a lazy-loaded 3D hero. It is a static site with no server of its own.

1. Core framework

Technology	Version	Role
React	19.2	UI library; component-based pages and sections
TypeScript	6.0	Static typing; <code>tsc -b</code> is the project's typecheck
Vite	8.2	Dev server and production bundler
React Router	7.18	Client-side routing (8 routes plus a 404 page)

2. Styling and design system

Technology	Version	Role
Tailwind CSS	4.3	Utility-first styling, via the <code>@tailwindcss/vite</code> plugin
Design tokens	—	<code>src/styles/tokens.css</code> is the single source of truth for colour, type scale and motion. Light and dark themes are supported.
Google Fonts	—	Inter Tight (headings), Inter (body), JetBrains Mono (labels)

Brand colours (sampled from the official logo): ■ Blue #3D47D5 ■ Crimson #9D1A40 ■ Ground #EFF0FE ■ Ink #0B0D1A

3. Animation and motion

Technology	Version	Role
GSAP + ScrollTrigger	3.15	Scroll-driven reveals, pinned sections, text animation
Lenis	1.3	Smooth scrolling. Navigation goes through <code>scrollToTop()</code> / <code>scrollToAnchor()</code> in <code>src/lib/motion.ts</code> .
Reduced-motion support	—	Animations respect the user's system preference

4. 3D graphics

Technology	Version	Role
Three.js	0.185	3D engine for the "Digital Healthcare Core" hero
React Three Fiber	9.7	React renderer for Three.js
Drei	10.7	Helpers for React Three Fiber
@react-three/postprocessing	3.1	Bloom and other post effects

The hero canvas is lazy-loaded so Three.js stays off the critical path: about 136 kB gzip initial load, with the 3D code deferred to the homepage.

5. Contact form and email

Technology	Role
Web3Forms	Serverless form backend. The form POSTs JSON to <code>api.web3forms.com/submit</code> and Web3Forms emails the enquiry to the company inbox. The visitor's email is the reply-to address, so the team replies from Gmail.
Access key	Public by design. It only routes to the registered inbox and lives in <code>src/data/content.ts</code> .

6. Tooling and quality

Tool	Role
oxlint 1.79	Linting (<code>npm run lint</code>)
Node scripts	<code>npm test</code> runs small <code>.mjs</code> checks for hero captions, count-up and the curtain latch
@vitejs/plugin-react 6.1	React support in Vite

```
npm run dev      # local dev server
npm run build    # tsc -b && vite build
npm run lint     # oxlint
npm test        # node test scripts
```

7. Hosting and deployment

Item	Detail
Live host	GitHub Pages, published by GitHub Actions (<code>.github/workflows/deploy.yml</code>) on every push to main
CI pipeline	Node 22 → <code>npm ci</code> → lint → build → copy <code>index.html</code> to <code>404.html</code> → deploy
Subpath handling	<code>vite.config.ts</code> sets base from the <code>GITHUB_PAGES</code> env var; the router basename comes from <code>import.meta.env.BASE_URL</code>
Deep links	Rewrites committed for every host: <code>404.html</code> (Pages), <code>vercel.json</code> , <code>public/_redirects</code> (Netlify / Cloudflare Pages)
Live URL	<code>talhahassan1998.github.io/valentiatechnologiesqatar</code>

8. Project structure

```
src/
├─ styles/tokens.css    brand tokens (colour, type, motion)
├─ lib/                 motion.ts (Lenis + GSAP), theme.ts, vGeometry.ts
├─ pages/               one component per route
├─ components/
│   └─ hero/            3D hero (code-split)
│       └─ sections/    composable page sections
│           └─ ui/       nav, footer, shared primitives
└─ data/content.ts      all site copy and contact details
```

9. Site map

Home · Solutions · Sectors · Services · Clients · Partners · About · Contact

Assets: the network texture and V-lattice images were generated with Higgsfield (Recraft V4.1) using the brand hexes as the palette.